

The two-parameter Hall–Littlewood exchange relation, and what actually lives on the hyperbola $st = 1$

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Abstract

Let $H_t(z)$ be the Hall–Littlewood vertex operator, $H_t(z)f[X] = \sigma[Xz]f[X - (1-t)/z]$. We prove that for two *independent* parameters s, t ,

$$(z_1 - tz_2)H_t(z_1)H_s(z_2) = (sz_1 - z_2)H_s(z_2)H_t(z_1),$$

by showing that both sides equal $(z_1 - z_2)\Phi$ for one and the same normal-ordered series Φ . Specialising $s = t$ recovers Zabrocki’s relation (2.14). The exchange factor $(sz_1 - z_2)/(z_1 - tz_2)$ is a scalar exactly on the hyperbola $st = 1$, where it equals t^{-1} ; the tempting conclusion is that the modes then generate a quantum torus. *They do not.* We compute the obstruction exactly: for $s = t^{-1}$ and all $m, n \in \mathbb{Z}$,

$$H_m^t H_n^{1/t} - t^{-1} H_n^{1/t} H_m^t = (1 - t^{-1}) t^{-m} h_{m+n}[(1+t)X],$$

a *multiplication* operator. It vanishes identically if and only if $t = 1$, where the vertex operator degenerates to multiplication by $\sigma[Xz]$ and the statement is trivial. So the hyperbola carries no quantum-torus structure; what it carries is the statement that the obstruction is as simple as an operator can be without being zero. Along the way we correct two errors in the brief that generated this problem.

1 Setup

Let $\Lambda = \Lambda_{\mathbb{Q}(t,s)}$ be the ring of symmetric functions over $\mathbb{Q}(t, s)$, with power sums $p_1, p_2, \dots$, complete homogeneous h_n ($h_n = 0$ for $n < 0$, $h_0 = 1$), and $\sigma[Xz] = \sum_{n \geq 0} h_n[X]z^n$. Plethystic notation is as usual: an expression $f[X + A]$ means substitution $p_k \mapsto p_k + p_k[A]$.

Definition 1.1. For a parameter u , the Hall–Littlewood vertex operator is

$$H_u(z)f[X] = \sigma[Xz] \cdot f\left[X - \frac{1-u}{z}\right], \quad H_u(z) = \sum_{m \in \mathbb{Z}} z^m H_m^u.$$

Lemma 1.2 (Modes are well defined). *Let $f \in \Lambda$ be of degree d . Then $f[X - (1-u)/z]$ is a Laurent polynomial in z^{-1} of degree at most d with coefficients in Λ , so*

$$f\left[X - \frac{1-u}{z}\right] = \sum_{j=0}^d c_j(f) z^{-j}, \quad H_m^u f = \sum_{j=0}^d c_j(f) h_{m+j}.$$

In particular $H_m^u f = 0$ for $m < -d$, and $H_m^u f$ is homogeneous of degree $d + m$.

Proof. Write f in the power-sum basis. For a monomial $p_\lambda = \prod_i p_{\lambda_i}$ the substitution $p_k \mapsto p_k - (1 - u^k)z^{-k}$ gives $\prod_i (p_{\lambda_i} - (1 - u^{\lambda_i})z^{-\lambda_i})$, which on expansion over subsets S of the parts is $\sum_S \prod_{i \in S} (-(1 - u^{\lambda_i})) p_{\lambda \setminus S} z^{-|\lambda_S|}$, a Laurent polynomial in z^{-1} of degree $\leq |\lambda| \leq d$. Multiplying by $\sigma[Xz] = \sum_{n \geq 0} h_n z^n$ and extracting z^m gives the stated formula, and the sum is finite. Since $h_{m+j} = 0$ unless $m + j \geq 0$ and $j \leq d$, we get $H_m^u f = 0$ for $m < -d$. $\square$

Consequently, for fixed f the double series $H_t(z_1)H_s(z_2)f$ and $H_s(z_2)H_t(z_1)f$ are well-defined elements of $\Lambda[[z_1^\pm, z_2^\pm]]$ with each coefficient a *finite* computation: their supports are the cones

$$\text{supp } H_t(z_1)H_s(z_2)f \subseteq \{b \geq -d, a + b \geq -d\}, \quad \text{supp } H_s(z_2)H_t(z_1)f \subseteq \{a \geq -d, a + b \geq -d\}, \quad (1)$$

where (a, b) is the exponent of $z_1^a z_2^b$. **The two cones are different.** This is the whole source of the phenomenon studied in §3.

We write ι_{12} for the expansion of a rational function of z_1, z_2 as a formal series in z_2/z_1 , and ι_{21} for the expansion in z_1/z_2 . Both are ring homomorphisms on the localisation of $\mathbb{Q}(t, s)[z_1^\pm, z_2^\pm]$ at the relevant denominators.

2 The two-parameter exchange relation

Lemma 2.1 (Contraction). *For a single letter v , $\sigma[-(1 - u)v] = \frac{1 - v}{1 - uv}$.*

Proof. $\sigma[A] = \exp(\sum_{k \geq 1} p_k[A]/k)$. For $A = -(1 - u)v$ we have $p_k[A] = -(1 - u^k)v^k$, so

$$\sigma[A] = \exp\left(\sum_{k \geq 1} \frac{u^k v^k - v^k}{k}\right) = \exp(-\log(1 - uv) + \log(1 - v)) = \frac{1 - v}{1 - uv}. \quad \square$$

Remark 2.2. The brief that generated this problem asserted $\sigma[-(1 - t)z_2/z_1] = \frac{1 - tz_2/z_1}{1 - z_2/z_1}$, which is the *reciprocal* of the truth. Followed literally, that sketch produces the relation with the two factors on opposite sides, i.e. $(sz_1 - z_2)H_t(z_1)H_s(z_2) = (z_1 - tz_2)H_s(z_2)H_t(z_1)$, which is false (§4, planted-error control: it fails in 12/12 tested cases).

Define the *normal-ordered series*

$$\Phi(z_1, z_2)f := \sigma[Xz_1]\sigma[Xz_2]f\left[X - \frac{1-t}{z_1} - \frac{1-s}{z_2}\right]. \quad (2)$$

Note that the argument of f is symmetric under $(z_1, t) \leftrightarrow (z_2, s)$: all of the asymmetry between the two orderings will be a scalar.

Lemma 2.3 (Both orders normal-order to the same Φ). *As identities of formal series with coefficients in $\text{End}(\Lambda)$,*

$$H_t(z_1)H_s(z_2) = \iota_{12}\left[\frac{z_1 - z_2}{z_1 - tz_2}\right]\Phi, \quad H_s(z_2)H_t(z_1) = \iota_{21}\left[\frac{z_1 - z_2}{sz_1 - z_2}\right]\Phi.$$

Proof. For the first,

$$\begin{aligned} H_t(z_1)H_s(z_2)f &= \sigma[Xz_1] \cdot \left(\sigma[Xz_2]f\left[X - \frac{1-s}{z_2}\right]\right)\left[X - \frac{1-t}{z_1}\right] \\ &= \sigma[Xz_1] \cdot \sigma\left[\left(X - \frac{1-t}{z_1}\right)z_2\right] \cdot f\left[X - \frac{1-t}{z_1} - \frac{1-s}{z_2}\right] \\ &= \sigma[Xz_1]\sigma[Xz_2] \cdot \sigma\left[-\frac{(1-t)z_2}{z_1}\right] \cdot f\left[X - \frac{1-t}{z_1} - \frac{1-s}{z_2}\right], \end{aligned}$$

using $\sigma[A + B] = \sigma[A]\sigma[B]$. By Lemma 2.1 with $v = z_2/z_1$ and $u = t$, the scalar factor is $\frac{1-z_2/z_1}{1-tz_2/z_1} = \frac{z_1-z_2}{z_1-tz_2}$. It arises from substituting $X \mapsto X - (1-t)/z_1$ into $\sigma[Xz_2] = \sum_{n \geq 0} h_n[X]z_2^n$, hence is produced as a power series in z_2/z_1 : that is the expansion ι_{12} .

For the second, the same computation with $(z_1, t) \leftrightarrow (z_2, s)$ gives the scalar $\sigma[-(1-s)z_1/z_2] = \frac{1-z_1/z_2}{1-sz_1/z_2} = \frac{z_1-z_2}{sz_1-z_2}$, produced as a power series in z_1/z_2 , i.e. ι_{21} . The function of f and the two σ 's are identical in both computations because the argument of f in (2) is symmetric. $\square$

Theorem 2.4 (Q99a). *For independent parameters s, t ,*

$$(z_1 - tz_2) H_t(z_1) H_s(z_2) = (z_1 - z_2) \Phi = (sz_1 - z_2) H_s(z_2) H_t(z_1).$$

Equivalently, at the level of modes, for all $m, n \in \mathbb{Z}$:

$$H_m^t H_n^s - t H_{m+1}^t H_{n-1}^s = s H_n^s H_m^t - H_{n-1}^s H_{m+1}^t. \quad (3)$$

Proof. Since ι_{12} is a ring homomorphism and $z_1 - tz_2$ is a unit times the denominator,

$$(z_1 - tz_2) \cdot \iota_{12} \left[\frac{z_1 - z_2}{z_1 - tz_2} \right] = \iota_{12} \left[(z_1 - tz_2) \cdot \frac{z_1 - z_2}{z_1 - tz_2} \right] = \iota_{12}[z_1 - z_2] = z_1 - z_2,$$

and identically $(sz_1 - z_2) \cdot \iota_{21} \left[\frac{z_1 - z_2}{sz_1 - z_2} \right] = z_1 - z_2$. Multiply the two identities of Lemma 2.3 by $(z_1 - tz_2)$ and $(sz_1 - z_2)$ respectively. Both give $(z_1 - z_2)\Phi$. Extracting the coefficient of $z_1^{m+1} z_2^n$ from the outer identity gives (3); each coefficient is a finite sum by Lemma 1.2, so no convergence question arises. $\square$

Corollary 2.5 (Zabrocki (2.14)). *Setting $s = t$ gives $(z_1 - tz_2)H(z_1)H(z_2) = (tz_1 - z_2)H(z_2)H(z_1)$, i.e. $H_m H_n - t H_{m+1} H_{n-1} = t H_n H_m - H_{n-1} H_{m+1}$.*

This is the relation recorded in M. Zabrocki, *On the action of the Hall–Littlewood vertex operator*, UCSD thesis, §2.5, eq. (2.14), read at source; and recorded — as known and attributed there to earlier work, not derived independently — in Korff [arXiv:1906.02565](#) (comp.tex ll. 753–759, “the latter are *known* to obey”) and Jing–Liu [arXiv:2303.10664](#) (eq. (e:com1), rearranged, with the $(1 - t^n)$ on the creation rather than the annihilation side, attributed to Jing 1991). A fourth reference carried by the brief, Necoechea–Rozhkovskaya [arXiv:1902.10049](#), does *not* contain this relation: its only t -deformed exchange relation is the generalized-fermion one for the twisted fields $\Psi^\pm$ on the charged Fock space, which is inequivalent. Its recovery from Theorem 2.4 is an external consistency check on the whole derivation.

Proposition 2.6 (Where the factor is a scalar). $\frac{sz_1 - z_2}{z_1 - tz_2}$ *is independent of (z_1, z_2) if and only if $st = 1$, in which case it equals t^{-1} .*

Proof. $sz_1 - z_2 = c(z_1 - tz_2)$ for a scalar c forces $c = s$ and $ct = 1$, i.e. $s = c = t^{-1}$. Conversely if $s = t^{-1}$ then $sz_1 - z_2 = t^{-1}(z_1 - tz_2)$. $\square$

3 The hyperbola $st = 1$: the defect is not zero

Fix $s = t^{-1}$, $t \neq 0$. Proposition 2.6 and Theorem 2.4 give

$$(z_1 - tz_2) \left(H_t(z_1) H_{1/t}(z_2) - t^{-1} H_{1/t}(z_2) H_t(z_1) \right) = 0,$$

and it is tempting to divide by $(z_1 - tz_2)$ and conclude the quantum-torus relation $H_m^t H_n^{1/t} = t^{-1} H_n^{1/t} H_m^t$. **This division is illegitimate**, because by (1) the difference is supported in a region on which $(z_1 - tz_2)$ is a zero divisor: the series $\delta(tz_2/z_1) = \sum_{j \in \mathbb{Z}} t^j z_1^{-j} z_2^j$ satisfies $(z_1 - tz_2)\delta(tz_2/z_1) = 0$. We now compute the defect exactly.

Notation 3.1. For $N \geq 0$ set $G_N := \sum_{a+b=N, a,b \geq 0} t^a h_a h_b$, and $G_N := 0$ for $N < 0$. Equivalently $G_N = h_N[(1+t)X]$, where $(1+t)X$ denotes the alphabet $X \cup tX$, i.e. the plethystic substitution $p_k \mapsto (1+t^k)p_k$; indeed $\sum_N G_N z^N = \sigma[tXz]\sigma[Xz] = \sigma[(1+t)Xz]$.

Lemma 3.2 (The δ -term). $(\iota_{12} - \iota_{21}) \left[\frac{z_1 - z_2}{z_1 - tz_2} \right] = (1 - t^{-1}) \sum_{j \in \mathbb{Z}} t^j z_1^{-j} z_2^j$.

Proof. $\iota_{12} \left[\frac{1}{z_1 - tz_2} \right] = \sum_{k \geq 0} t^k z_2^k z_1^{-k-1}$ and $\iota_{21} \left[\frac{1}{z_1 - tz_2} \right] = - \sum_{k \geq 0} t^{-k-1} z_1^k z_2^{-k-1}$; their difference is $\sum_{j \in \mathbb{Z}} t^j z_1^{-j-1} z_2^j$ (the terms $j \geq 0$ from the first sum, the terms $j = -k-1 < 0$ from the second). Multiplying by the polynomial $z_1 - z_2$,

$$\sum_{j \in \mathbb{Z}} t^j z_1^{-j} z_2^j - \sum_{j \in \mathbb{Z}} t^j z_1^{-j-1} z_2^{j+1} = \sum_{j \in \mathbb{Z}} (t^j - t^{j-1}) z_1^{-j} z_2^j,$$

after reindexing $j \mapsto j-1$ in the second sum. $\square$

Lemma 3.3 (Diagonal cancellation). Let $s = t^{-1}$ and let $\Phi_{a,b}$ denote the coefficient of $z_1^a z_2^b$ in Φ . Then for every $N \in \mathbb{Z}$ and every $f \in \Lambda$, the sum $\sum_{l \in \mathbb{Z}} t^l \Phi_{l, N-l} f$ is finite and

$$\sum_{l \in \mathbb{Z}} t^l \Phi_{l, N-l} f = G_N \cdot f.$$

Proof. Expand f in the power-sum basis and apply the two-variable analogue of the expansion in Lemma 1.2: assigning each part of λ to “keep”, “ z_1 ” or “ z_2 ”,

$$f \left[X - \frac{1-t}{z_1} - \frac{1-s}{z_2} \right] = \sum_{(\mu, i, i')} c_{\mu, i, i'} p_\mu z_1^{-i} z_2^{-i'}, \quad i, i' \geq 0, i + i' \leq d.$$

Hence $\Phi_{a,b} f = \sum c_{\mu, i, i'} p_\mu h_{a+i} h_{b+i'}$, which vanishes unless $a \geq -d$ and $b \geq -d$; so the sum over l is finite. Substituting $a = l + i$,

$$\sum_l t^l \Phi_{l, N-l} f = \sum_{(\mu, i, i')} c_{\mu, i, i'} t^{-i} p_\mu \sum_{a \geq 0} t^a h_a h_{N+i+i'-a} = \sum_{w \geq 0} \left(\sum_{i+i'=w} c_{\mu, i, i'} t^{-i} \right) p_\mu G_{N+w}.$$

Now the inner bracket is exactly the coefficient of z_2^{-w} in the specialisation $z_1 \mapsto tz_2$ of the displayed expansion. For $s = t^{-1}$ that specialisation is

$$-\frac{1-t^k}{(tz_2)^k} - \frac{1-t^{-k}}{z_2^k} = \frac{-t^{-k} + 1 - 1 + t^{-k}}{z_2^k} = 0 \quad \text{for every } k \geq 1,$$

i.e. $f \left[X - \frac{1-t}{z_1} - \frac{1-s}{z_2} \right] \Big|_{z_1=tz_2} = f[X]$. Therefore the bracket vanishes for $w > 0$ and equals the coefficient f_μ of p_μ in f for $w = 0$. Only the $w = 0$ term survives, giving $\sum_\mu f_\mu p_\mu G_N = G_N \cdot f$. $\square$

Theorem 3.4 (Q99b). Let $s = t^{-1}$ with $t \neq 0$. Then for all $m, n \in \mathbb{Z}$, as operators on Λ ,

$$H_m^t H_n^{1/t} - t^{-1} H_n^{1/t} H_m^t = (1 - t^{-1}) t^{-m} h_{m+n} [(1+t)X].$$

where the right-hand side is the operator of multiplication by the symmetric function $h_{m+n}[(1+t)X] = G_{m+n}$ (which is 0 when $m+n < 0$). Equivalently, in generating-function form,

$$H_t(z_1)H_{1/t}(z_2) - t^{-1}H_{1/t}(z_2)H_t(z_1) = (1 - t^{-1}) \delta\left(\frac{tz_2}{z_1}\right) \sigma[(1+t)Xz_2] \cdot.$$

Proof. By Lemma 2.3 with $s = t^{-1}$ we have $\frac{z_1 - z_2}{sz_1 - z_2} = \frac{t(z_1 - z_2)}{z_1 - tz_2}$, hence $t^{-1}H_{1/t}(z_2)H_t(z_1) = \iota_{21}\left[\frac{z_1 - z_2}{z_1 - tz_2}\right]\Phi$, and so

$$C := H_t(z_1)H_{1/t}(z_2) - t^{-1}H_{1/t}(z_2)H_t(z_1) = (\iota_{12} - \iota_{21})\left[\frac{z_1 - z_2}{z_1 - tz_2}\right]\Phi.$$

By Lemma 3.2 this equals $(1 - t^{-1}) \sum_{j \in \mathbb{Z}} t^j z_1^{-j} z_2^j \Phi$. Extract the coefficient of $z_1^m z_2^n$: the pairs contributing are $(a, b) = (m + j, n - j)$, so

$$C_{m,n}f = (1 - t^{-1}) \sum_{j \in \mathbb{Z}} t^j \Phi_{m+j, n-j} f,$$

a finite sum because $\Phi_{a,b}f = 0$ unless $a \geq -d$ and $b \geq -d$ (Lemma 3.3), which confines j to $-d - m \leq j \leq n + d$. Writing $j = l - m$ and applying Lemma 3.3 with $N = m + n$,

$$C_{m,n}f = (1 - t^{-1}) t^{-m} \sum_{l \in \mathbb{Z}} t^l \Phi_{l, m+n-l} f = (1 - t^{-1}) t^{-m} G_{m+n} \cdot f. \quad \square$$

Corollary 3.5 (The quantum torus does *not* appear). *On the hyperbola $st = 1$ the relation $H_m^t H_n^{1/t} = t^{-1} H_n^{1/t} H_m^t$ holds for all m, n if and only if $t = 1$.*

Proof. By Theorem 3.4 the defect is $(1 - t^{-1})t^{-m}G_{m+n}$. Since $G_0 = 1$, taking $m + n = 0$ gives defect $(1 - t^{-1})t^{-m} \neq 0$ unless $t = 1$. Conversely at $t = 1$ the prefactor $1 - t^{-1}$ vanishes, so the defect is identically zero. $\square$

Remark 3.6 ($t = 1$ is degenerate, and the brief's negative control was wrong). The brief asserted: "At $t = s = 1$ the hyperbola is met, $1/t = 1$, and the naive corollary predicts that the Bernstein operators commute. *They do not* ... Any derivation of the mode relation that does not break at $t = 1$ is wrong."

This is false, and following it would have caused me to reject the correct theorem. At $t = 1$ Definition 1.1 degenerates: $H_1(z)f = \sigma[Xz]f[X - 0] = \sigma[Xz]f$, so H_m^1 is *multiplication by h_m* — verified independently in §4 — and multiplication operators commute. The brief's supporting observation, that (2.14) at $t = 1$ is strictly weaker than commutativity, is true but does not entail non-commutativity: commutativity at $t = 1$ is an extra true fact that (2.14) alone does not see. Corollary 3.5 is therefore consistent with the $t = 1$ specialisation, and the prediction does move: the prefactor $1 - t^{-1}$ takes the values 0, $\frac{1}{2}$, 2, -1 at $t = 1, 2, -1, \frac{1}{2}$.

Remark 3.7 (What the hyperbola actually carries). The positive content of Theorem 3.4 is not a relation but a *regularity*. For generic (s, t) the two orderings differ by a (z_1, z_2) -dependent factor and no mode-level statement of quantum-torus type can even be formulated. On $st = 1$ one can formulate it, and it fails — but it fails by a multiplication operator, the simplest kind of endomorphism of Λ there is. The mechanism is exactly the plethystic cancellation in Lemma 3.3: on the diagonal $z_1 = tz_2$, which is where the δ -term is supported, the two shifts $-\frac{1-t}{z_1}$ and $-\frac{1-t^{-1}}{z_2}$ annihilate each other, so f passes through untouched and only the two σ 's survive. The hyperbola is precisely the locus where the singular support of the exchange and the vanishing locus of the total plethystic shift coincide.

Remark 3.8 ($t = -1$, the free-fermion point). At $t = -1$ (the intersection of $st = 1$ with the diagonal $s = t$, other than $t = 1$) the substitution $p_k \mapsto (1 + (-1)^k)p_k$ kills all odd power sums, so $G_N = 0$ for N odd and $G_{2M} = h_M[X^2]$ (the alphabet of squares). The defect is therefore $2(-1)^m h_{(m+n)/2}[X^2]$ for $m + n$ even and 0 for $m + n$ odd. For $m = n = 1$ this is $-2p_2$, which is consistent with the known failure $H_1 H_1 \cdot 1 = -p_2 \neq 0$ of the naive anticommutation reading at $t = -1$.

4 Verification

All computations were done in an engine written for this session (`projects/scratch/q99/engine.py`): symmetric functions as dictionaries $\{\text{partition} \rightarrow \text{coefficient}\}$ in the power-sum basis, $h_n = \sum_{\lambda \vdash n} p_\lambda / z_\lambda$, and the plethystic shift implemented by explicit subset expansion. It shares no code path with the pre-existing scripts `/tmp/browse/0907pm/hlvo.py` and `hlvo2.py` (which use `sympy.subs`, `Poly`, and `exp(\sum p_k u^k / k)`); those were read but not run, and nothing here depends on them.

Engine anchors (untuned; neither was used to build the engine):

- $t = 1$: $H_m^1 f = h_m f$ for $m \in \{0, 1, 2, 3\}$ and $f \in \{1, p_1, p_1^2, p_2\}$ — 16/16.
- $t = 0$: the classical Bernstein operator, $H_m^0 s_\lambda = s_{(m, \lambda)}$ for $m \geq \lambda_1$, with s_λ computed independently by Jacobi–Trudi — 13/13; straightening ($m < \lambda_1$) also behaves correctly, e.g. $H_1^0 s_{(2)} = 0$ and $H_0^0 s_{(2)} = -s_{(1,1)}$.

Theorem 2.4 (mode form (3)), generic symbolic t, s : 126 cases = 6 arguments $f \in \{1, p_1, p_2, p_1^2, p_3, p_2 p_1\} \times 21$ pairs (m, n) including negative modes. Residual identically 0 in all 126.

Non-degeneracy. At $(m, n) = (1, 2)$, $f = p_1$ the four terms of (3) are pairwise distinct and each depends on both t and s ; the bare commutator $H_1^t H_2^s f - H_2^s H_1^t f$ is nonzero. The test is not vacuous.

Planted-error control. Replacing the relation by $(z_1 - \alpha z_2)H_t H_s = (\beta z_1 - \gamma z_2)H_s H_t$ for $(\alpha, \beta, \gamma) \in \{(s, t, 1), (t, t, 1), (t, 1/s, 1), (ts, s, 1), (t, s, s)\}$ gives a nonzero residual in 12/12 tested cases for each of the five perturbations; only $(t, s, 1)$ passes. In particular the inverted sketch of Remark 2.2 — which is the variant $(s, t, 1)$ — fails 12/12.

Lemma 2.3 (the proof, not just the statement): both $H_m^t H_n^s f - t H_{m+1}^t H_{n-1}^s f$ and $s H_n^s H_m^t f - H_{n-1}^s H_{m+1}^t f$ were checked against $\Phi_{m,n} f - \Phi_{m+1,n-1} f$ with Φ computed directly from (2): 65/65 agree for each ordering (5 arguments $\times$ 13 mode pairs).

Theorem 3.4, generic symbolic t with $s = t^{-1}$: 138 cases = 6 arguments $\times$ 23 pairs (m, n) , mismatches 0. The defect was *nonzero* in 120 of the 138; the 18 vanishing cases factor as 3×6 — the three pairs with $m + n < 0$, times the six arguments — exactly as Theorem 3.4 predicts, and for no other reason. A higher-degree stress test (arguments $p_2^2, p_4, p_2 p_1^2, p_3 p_2$ of degrees 4 and 5; modes up to 5 and down to -2) gives a further 28/28.

The two intermediate steps of the proof of Theorem 3.4 were checked separately, so that the final agreement is not the only evidence: $C_{m,n} = (1 - t^{-1}) \sum_j t^j \Phi_{m+j, n-j}$ (21/21) and Lemma 3.3 $\sum_l t^l \Phi_{l, N-l} = G_N$ (15/15). The two expressions for G_N ($\sum_{a+b=N} t^a h_a h_b$ and $h_N[(1+t)X]$) agree for $0 \leq N \leq 5$.

Controls on the hyperbola (Remark 3.6): at $t = 1$ the defect is identically 0 over all tested (m, n) and f ; at $t = 2$, $t = -1$ and $t = \frac{1}{2}$ it is nonzero, and matches Theorem 3.4 in every case. At $t = -1$ it vanishes exactly on $m + n$ odd, as predicted.

5 Relation to the ribbon-side $ts = 1$ locus, and what is *not* claimed

PROVE 2026-09-07 c2 (proofs/2026-09-07-c2-Q96-order-over-sublattice.tex, Thm 4.1(2), Cor 4.2(iii)) established, on two-bead matrix elements of the ribbon operators, that $\langle M' | [R_e(t), R_f(s)] | M \rangle = \sum_{(b,c)} t^{P-k} s^Q (1 - (ts)^k)$, hence that the *two-bead sector* vanishes identically iff $ts = 1$. It is only that sector: the one-bead sector of the same commutator is nonzero at every point of $ts = 1$ except $t = -1$ — verified in the review of 8 September 2026 on 1232/1232 entries ($|\lambda| \leq 6$, $1 \leq e, f \leq 4$), where it is moreover divisible by $(1 + t)$ on all of them. The hyperbola is thus named by two computations. Are they two witnesses or one witness twice?

They are not the same witness, and this session’s result is what distinguishes them:

- On the ribbon side the vanishing object is one *sector* of the commutator, not the commutator itself; and the two sides’ degenerate points along the hyperbola are exactly swapped. On $ts = 1$ the ribbon commutator $[R_e(t), R_f(1/t)]$ is nonzero at $t = 1$ (400 entries, $|\lambda| \leq 5$, $1 \leq e, f \leq 4$) and identically zero at $t = -1$; the twisted vertex-operator commutator is zero at $t = 1$ (Corollary 3.5) and nonzero at $t = -1$ ($-2p_2$ at $m = n = 1$, Remark 3.8). Complementary vanishing loci on the same hyperbola: no reparametrisation carries one derivation to the other.
- On the vertex-operator side the plain commutator does not vanish on $ts = 1$: e.g. $[H_1^t, H_1^{1/t}] \cdot 1 = \frac{t^2-1}{t} h_2$, which is nonzero for $t \neq \pm 1$, and $[H_2^t, H_1^{1/t}] \cdot 1 \neq 0$ at $t = -1$. The natural t^{-1} -twisted commutator does not vanish on $ts = 1$ either, by Corollary 3.5: it vanishes only at the single point $t = 1$.

So the two occurrences of $st = 1$ have different content. The vertex-operator occurrence is Proposition 2.6 — a statement about when a rational exchange factor degenerates to a scalar — and the mode-level consequence is a *failure* with a computed defect, not a vanishing.

Explicitly not claimed here. That $st = 1$ is “the hidden structure behind the ribbon commutator”; that $R_e(t)$ is a mode H_m^t or any specialisation of one; that Theorem 3.4 transports to the ribbon side. $R_e(t)$ is the connected truncation of the ribbon-move expansion and the Q92/Q96 enumeration runs over single bead moves, not over $\sigma[Xz]$; a transport would be a computation, and it was not done. Theorem 2.4 warrants Theorem 2.4.

6 Gaps and what was not attempted

1. **Not attempted: the transport to $R_e(t)$.** See §5. This is the natural next question and the obstruction is named there.
2. **Not attempted: a basis-level interpretation.** Theorem 3.4 says the defect is multiplication by $h_{m+n}[(1+t)X]$. The alphabet $(1+t)X$ is the one for which $\prod_i (1 - x_i z)^{-1} (1 - t x_i z)^{-1}$ is the generating function; it is natural to ask whether the defect is the image of a Hall–Littlewood-theoretic operator (a Macdonald q, t specialisation, or the t -analogue of a Boson–Fermion normal-ordering constant). Not investigated.
3. **Coverage.** Theorem 2.4 and Theorem 3.4 are proved for all $m, n \in \mathbb{Z}$ and all $f \in \Lambda$; the computations are checks, not the proof. The checks cover arguments of degree ≤ 5 and modes in $[-2, 5]$. No claim is made about specialisations where $t = 0$ (Definition 1.1 still makes sense, but $s = t^{-1}$ does not).

4. **The ι formalism.** Lemma 2.3 identifies which expansion region each ordering produces. I have argued this from the derivation (the factor arises by substituting into $\sigma[Xz_2] = \sum_{n \geq 0} h_n z_2^n$, a power series in z_2); the support statement (1), proved from Lemma 1.2, is the independent confirmation, and the 65/65 numerical agreement in §4 is a third. I regard this as closed but flag it as the one step where an expansion-region error would be invisible in the final formula.